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Highly efficient synthesis of di-substituted allenes via copper-catalyzed S_N2'- substitution reaction of organotitanium reagents with propargyl bromides

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ABSTRACT

A simple and mild catalytic S_N2'-type displacement of propargyl halides with organotitanium reagents is reported. The S_N2'-type displacement of propargyl bromides with organotitanium reagents mediated by CuI (2 mol %)/$MePPh_2$(4 mol%) in dichloro methane afforded multi-substituted allenes in good yields (up to 95 %) at 40 °C for 5 h. The aryls bearing electron-donating or electron-withdrawing groups in aryltitanium reagents gave multi-substituted allenes in good yields. In addition, the di-substituted allenes bearing a naphthyl group was obtained in 54–99 % selectivities with isolated yields of 43–83 %. The process was simple and easily performed, and it provides an efficient method for the synthesis of di-substituted allene derivatives. On the basis of the experimental results, a possible catalytic cycle has been proposed.

1. Introduction

Allenes are a kind of very useful organic compounds. As important synthetic fragments and functional groups, allenes have been widely applied for many transformations in organic synthesis [1,2]. Allene moieties have also been found in many of natural products, pharmaceuticals [3], and optical materials [4]. Due to the wide range of applications of allenes, the synthesis of allenes has attracted great attention of many organic chemists and medical chemists. Developing efficient reactions for the synthesis of allenes from simple and readily available organic compounds is very important. Based on various utilities of allenes, many effective synthetic methodologies of synthesis allene compounds have been reported [5,6]. Typical synthetic protocols for substituted allenes include elimination of allylic compounds [7], isomerization of alkynes [8], a reaction of aldehydes and terminal alkynes [9], and S_N2'-type displacement of propargylic compounds [10]. Among these methods hitherto developed, the metal-catalyzed S_N2'-type displacement of propargyl alcohol derivatives with organometal species is one of the most generally useful one (Scheme 1). However, there are two competing possible reaction pathways. The successful application of such a substitution reaction relies on the choice of catalyst and/or organometallic reagents to selectively produce either allene product **2** or alkyne product **3**. Therefore, the development of rapid and more efficient procedures for the synthesis of allenes still remains a challenge.

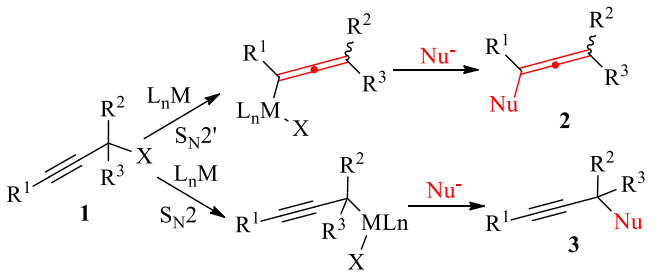
Our previous studies show that organotitanium reagents are highly efficient nucleophiles for cross-coupling reactions with organic halides [11], and the investigations have demonstrated that copper is a good catalytic metal [12]. To continue our effort to develop coupling reactions using reactive organometallic reagents (6r,6s [12-13]), we herein report a CuI (2 mol%)/$MePPh_2$ (4 mol%) catalyzed S_N2'-type displacement of propargyl halides with organotitanium reagents at 40 °C in short reaction time with good yields for the synthesis of allenes. The process was simple and easily performed, and it provides an efficient method for the synthesis of substituted allene derivatives.

2. Results and discussion

To optimize the reaction conditions, effects of copper source, ligand, solvent, reaction time, the amount of organotitanium, and the molar ratio of metal to ligand on the S_N2' substitution reaction were investigated using 1-bromopent-2-yne (**1a**) and phenyl titanium (**2a**) as reaction substrates (Table 1 and Table 2). In the preliminary study, the formation of pentane-1,2-diene-3-ylbenzene (**3aa**) and pentane-2-yne-1-ylbenzene (**4aa**) were not detected when 1-bromopent-2-yne (**1a**) was

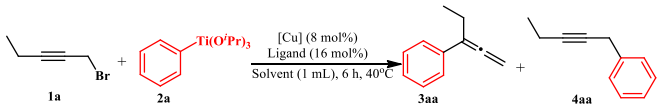
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Scheme 1. S_N2' and S_N2 processes of metal-catalyzed coupling reactions of propargyl derivatives with organometallic nucleophiles.

Table 1
Effect of the copper source, ligands and solvents on the S_N2' substitution reaction.[a]

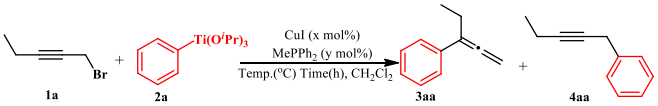


Entry	Cu. Salt (8 mol %)	Ligand (16 mol %)	Solvent	3aa: 4aa (%)[b]	Yield 3aa (%)[c]
1	–	–	CH_2Cl_2	–	trace
2	CuI	–	CH_2Cl_2	–	trace
3	CuI	Bipyrid	CH_2Cl_2	93:7	72
4	$CuCl_2$	Bipyrid	CH_2Cl_2	93:7	43
5	CuCl	Bipyrid	CH_2Cl_2	93:7	65
6	CuBr	Bipyrid	CH_2Cl_2	93:7	64
7	$Cu(OAc)_2$	Bipyrid	CH_2Cl_2	93:7	63
8	$Cu(acac)_2$	Bipyrid	CH_2Cl_2	93:7	65
9	CuI	1,10-phen	CH_2Cl_2	94:6	66
10	CuI	PPh_3	CH_2Cl_2	93:7	68
11	CuI	PCy_3	CH_2Cl_2	95:5	75
12	CuI	$MePPh_2$	CH_2Cl_2	98:2	77
13	CuI	$(4\text{-}MeC_6H_4)_3P$	CH_2Cl_2	96:4	73
14	CuI	$MePPh_2$	THF	96:4	68
15	CuI	$MePPh_2$	Toluene	–	39
16	CuI	$MePPh_2$	Hexane	–	18

[a] **1a**/**2a**/[Cu]/Ligand = 0.5/0.75/0.04/0.08 mmol.
[b] The ratio of **3aa**/**4aa** was determined by isolated yield. "-"has not been calculated.
[c] Yield of **3aa** was isolated.

treated with phenyltitanium (**2a**) in THF at 40 °C for 3 h without any metal or ligand (Table 1, entry 1). When CuI was used as a catalyst, the conversion rate of the coupling reaction between 1-bromopent-2-yne (**1a**) and phenyltitanium (**2a**) is also poor (Table 1, entry 2). Interestingly, the CuI-catalyzed the S_N2' substitution reaction of 1-bromopent-2-yne (**1a**) with phenyl titanium (**2a**) in the presence of 16 mol % bipyridine, the pentane-1,2-diene-3-ylbenzene (**3a**) was obtained in 72 % isolated yield with 93:7 selectivity (Table 1, entry 3). Other copper sources with bipyridine were subsequently surveyed. Although $CuCl_2$, CuCl, CuBr, $Cu(OAc)_2$, and $Cu(acac)_2$ can effectively catalyzed the S_N2' substitution reaction of 1-bromopent-2-yne (**1a**) with phenyltitanium (**2a**), the selectivities and isolated yields of allene (**3aa**) were lower than that of CuI (Table 1, entries 4–8). The other ligands were further examined (Table 1, entries 9–13). It was found that the $MePPh_2$ was the best effective ligand for the reactivity and selectivity (Table 1, entry 12). Other ligands did not provide satisfactory results. Then, various solvents were screened under the model reaction conditions, and the results are summarized in Table 1 (entries 14–16). Solvents such as THF, toluene, and hexane were unsuitable for this reaction since they gave lower yields and selectivities (Table 1, entries 14–16). In contrast, when CH_2Cl_2 was used as solvent, allene (**3aa**) was obtained in 77 % isolated yield with 98:2 selectivity (Table 1, entry 12).

Table 2
Effect of the reaction temperature and time, the loading of CuI and $PhTi(O^iPr)_3$ on the S_N2' substitution reaction.[a]



Entry	CuI (x mol %)	$MePPh_2$ (y mol %)	PhTi $(O^iPr)_3$ (equiv.)	3 : 4 (%)[b]	3aa (%)[b]
1	8	24	1.5	95:5	60
2	8	8	1.5	99:1	74
3	10	20	1.5	99:1	74
4	4	8	1.5	99:1	76
5	2	4	1.5	>99:1	79
6	2	4	1.2	90:10	69
7	2	4	1.8	99:1	78
8[d]	2	4	1.5	96:4	64
9[e]	2	4	1.5	99:1	79
10[f]	2	4	1.5	99:1	79
11[g]	2	4	1.5	90:10	64
12[h]	2	4	1.5	>99:1	79

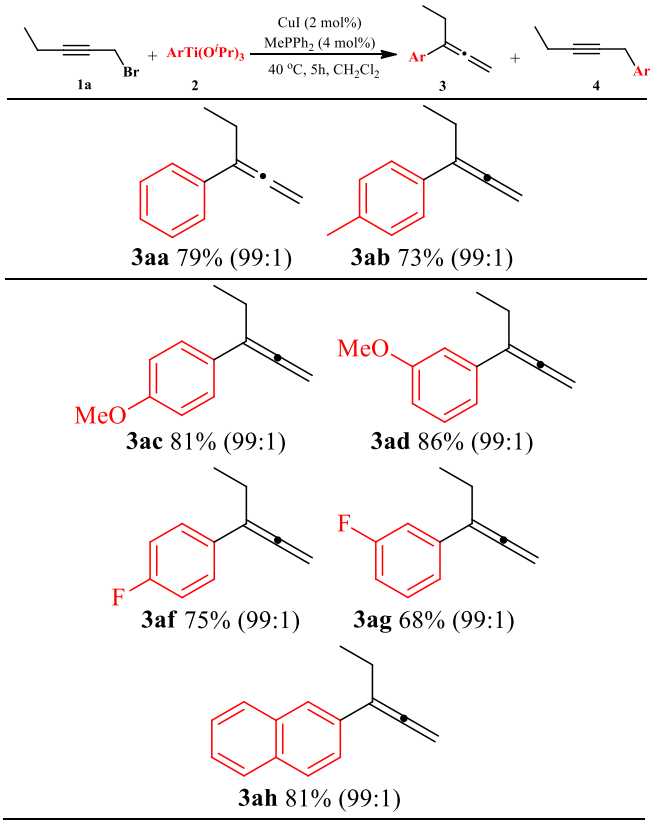
[a] **1a** = 0.5 mmol, CH_2Cl_2, 40 °C, 5 h.
bYield of **3aa** was isolated.
c The ratio of **3aa**/**4aa** was isolated by isolated yield.
d4 h, 40 °C.
e5 h, 40 °C.
f7 h, 40 °C.
[g]5 h, 25 °C.
[h]5 h, 50 °C.

To further study the reactivity and selectivity, the reaction of bromopent-2-yne (**1a**) with phenyltitanium (**2a**) under other conditions were investigated. When the ratio of CuI and $MePPh_2$ was altered to 1:1 or 1:3 from 1:2, low isolate yield of allene (**3aa**) was obtained (Table 2, entries 1 and 2). When the CuI loading was increased from 8 mol% to 10 mol%, the isolated yield of allene (**3aa**) decreased from 77 % to 74 % (Table 1, entry 12; Table 2, entry 3). When the CuI loading was decreased from 8 mol% to 4 mol%, the isolated yield of allene (**3aa**) almost unchange (Table 1, entry 12; Table 2, entry 4). While, excellent selectivity (**3aa**:**4aa**>99:1) and good isolated yield of allene (**3aa**) (79 %) were obtained when the CuI loading was decreased from 8 mol% to 2 mol% (Table 1, entry 12; Table 2, entry 5). This may be due to the reduced amount of CuI accelerating the S_N2' substitution reaction while inhibiting S_N2 substitution reaction. The effect of the amount of PhTi $(O^iPr)_3$ was also investigated. When the $PhTi(O^iPr)_3$ loading was decreased from 0.75 mmol to 0.6 mmol, the isolated yield of allene (**3aa**) decreased from 79 % to 69 % (Table 2, entries 5 and 6). However, when the $PhTi(O^iPr)_3$ loading was increased from 0.5 mmol to 0.9 mmol, the isolated yield of allene (**3aa**) remained unchange (Table 2, entries 5 and 7). Subsequently, the effect of the reaction time and temperature were also investigated. When the reaction time was increased from 6 h to 7 h or decreased from 6 h to 5 h, the isolated yield of allene (**3aa**) remained unchanged (Table 2, entries 5, 9 and 10). While the reaction time was decreased from 6 h to 4 h, the isolated yield of allene (**3aa**) decreased from 79 % to 64 % (Table 2, entries 5, 8). When the temperature was decreased from 40 °C to 25 °C, the isolated yield of allene (**3aa**) decreased from 79 % to 64 % (Table 2, entries 9, 11). However, the isolated yield of allene (**3aa**) remained unchanged when the temperature was increased from 40 °C to 50 °C (Table 2, entries 9 and 12).

Therefore, the optimal S_N2' substitution reaction conditions were 2 mol % CuI, 4 mol % $MePPh_2$, 0.75 mmol $PhTi(O^iPr)_3$, 0.5 mmol propargyl bromide in CH_2Cl_2 (1 mL) at 40 °C for 5 h.

With the optimized conditions in hand, the scope of catalytic S_N2' substitution reaction of different aryltitanium reagents (**2**) with bromopent-2-yne (**1a**) was then explored, and results were presented in

Table 3
CuI/$MePPh_2$-catalyzed the S_N2' substitute reaction of bromopent-2-yne (**1a**) with aryltitanium reagents (**2**).[a],[b],[c]

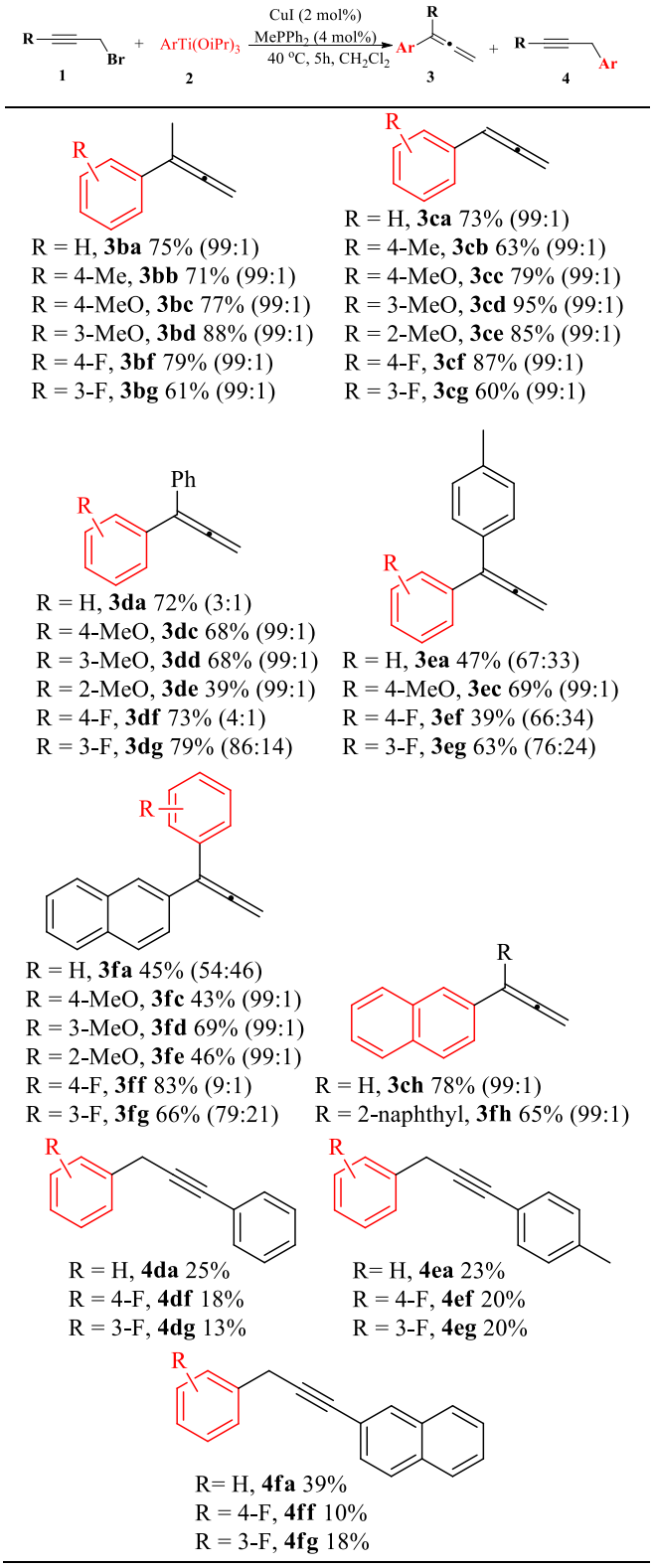


[a] **1a**/$ArTi(O^iPr)_3$/CuI/$MePPh_2$ = 0.5/0.75/0.01/0.02 mmol.
[b] The ratio of **3**/**4** was determined by isolated yield.
[c] Isolated yield, two runs.

Table 3. The corresponding product allenes (i.e., **3aa**–**3ad**, **3af-3ah**) were obtained in good yields with high selectivities. Reactions of bromopent-2-yne (**1a**) with phenyl titanium (**2a**) furnished penta-1,2-dien-3-ylbenzene (**3aa**) with 99 % selectivity in isolated yield of 79 % (Table 3). The reactions of aryltitanium reagents with electron-donating (**2b-2d**) or electron-withdrawing groups (**2f**, **2g**) on the benzene rings underwent the S_N2' substitution reactions smoothly to give the di-substituted allenes with moderate to good yields (68–86 %, Table 3, **3 ab-3ad**, **3af**, **3 ag**). Under the same conditions, 2-naphthyl titanium (**2h**) can also reacted smoothly with bromopent-2-yne (**1a**) to provide the 2-(1-(4-fluorophenyl)propa-1,2-dien-1-yl) naphthalene (**3ah**) in 81 % isolated yield with good selectivity (Table 3).

Encouraged by the good performance of the current catalyst system shown above, we subsequently investigated the S_N2' substitution reaction of different propargyl bromides **1** (**b-f**) with aryltitanium reagents **2** (**a-h**). The results were summarized in Table 4. In all the cases, 39–95 % isolated yields of di-substitute allenes were obtained for all evaluated substrates (Table 4). The results indicated that the reactions of aryltitanium reagents bearing electron-donating or electron-withdrawing groups on the benzene rings with aliphatic propargyl bromides (such as 1-bromobut-2-yne (**1b**), 3-bromoprop-1-yne (**1c**)) provided the desired di-substituted allene products in good isolated yields (up to 95 %) with high selectivities (up to 99 %) (Table 4, **3bb-3bd**, **3bf**, **3bg**, **3 cb-3cg**, **3ch**). Besides aliphatic propargyl bromides, aromatic propargyl bromides (**1d-1f**) were also found to be suitable substrates for the S_N2' substitute reactions, and the reaction provided the desired di-substituted allene products in good isolated yields (up to 79 %) with high selectivities (up to 99 %) (Table 4, **3da-3dg**, **3ea**, **3ec**, **3ef**, **3eg**). Aromatic

Table 4
CuI/$MePPh_2$-catalyzed the S_N2' substitute reaction of propargyl bromides (**1**) with aryltitanium reagents (**2**).[a],[b],[c]



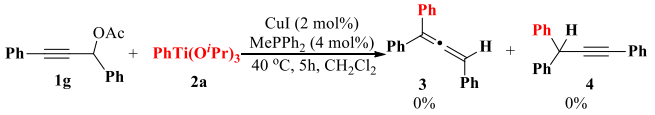
[a] **4**/$ArTi(O^iPr)_3$/CuI/$MePPh_2$ = 0.5/0.75/0.01/0.02 mmol.
[b] The ratio of **3**/**4** was determined by isolated yield.
[c] Isolated yield, two runs.

propargyl bromides containing 2-naphthyl group also produced the allene products with 43–83 % isolated yields (Table 4, **3fa-3fh**). The S_N2' substitute reaction of (3-bromoprop-1-yn-1-yl)benzene (**1d**) with phenyl titanium (**2a**) was explored, and after 5 h, allene product **3da** was formed with 75 % selectivity in 72 % isolated yield. The S_N2' substitute reaction of 3-flourophenyltitanium (**2g**) with (3-bromo prop-1-yn-1-yl) benzene (**1d**) was also explored, the reaction proceeded smoothly to give the products **3 dg** and **4 dg** with a ratio of 86:14 in favor of **3 dg** and the allene product **3 dg** with 79 % isolated yield. The S_N2' substitute reaction of 4-flourophenyl titanium (**2f**) with (3-bromoprop-1-yn-1-yl) benzene (**1d**) proceeded smoothly to give the products **3df** and **4df** with a ratio of 4:1 in favor of **3df** and the product **3df** with 73 % isolated yield. Although 39–69 % isolated yields of di-substitute allenes (Table 4, **3ea, 3ef, 3fh**) were obtained, the S_N2' substitute reactions of 1-(3-bromo prop-1-yn-1-yl)-4-methyl benzene (**1e**) with aryltitaniums (**2a**,**2f**,**2g**) gave low ratio of **3**:**4** (Table 4, 66:34–76:24). The S_N2' substitute reactions of 2-(3-bromoprop-1-yn-1-yl)naphthalene (**1f**) with aryltitaniums (**2c**,**2d**-**2f**) proceeded smoothly to give the di-substitute allenes in 43–83 % isolated yields with good selectivities (90:10 to 99:1) (Table 4, **3fc-3ff**). The S_N2' substitute reaction of 2-(3-bromoprop-1-yn-1-yl) naphthalene (**1f**) with 3-flourophenyl titanium (**2g**) gave the products **3 fg** and **4 fg** with a ratio of 79:21 in favor of **3 fg** and the allene product **3 fg** with 66 % isolated yield. However, under the same reaction conditions, the S_N2' substitute reaction of 2-(3-bromoprop-1-yn-1-yl)naphthalene (**1f**) and phenyl titanium (**2a**) gave the products **3fa** and **4fa** in a ratio of 54:46 only, which was slightly favorable for **3fa**. However, the isolated yield of the allene product **3fa** was 45 % only, while the alkyne **4fa** was obtained with an isolated of 39 % (Table 4). This may be due to the steric hindrance of propargyl bromides, which hinders the S_N2' substitution reaction. Pleasingly, the S_N2' substitute reaction of 2-(3-bromoprop-1-yn-1-yl)naphthalene (**1f**) with 2-naphthyltitanium (**2h**) proceeded smoothly and gave the allene product **3fh** in 65 % isolated yield with high selectivity (99:1).

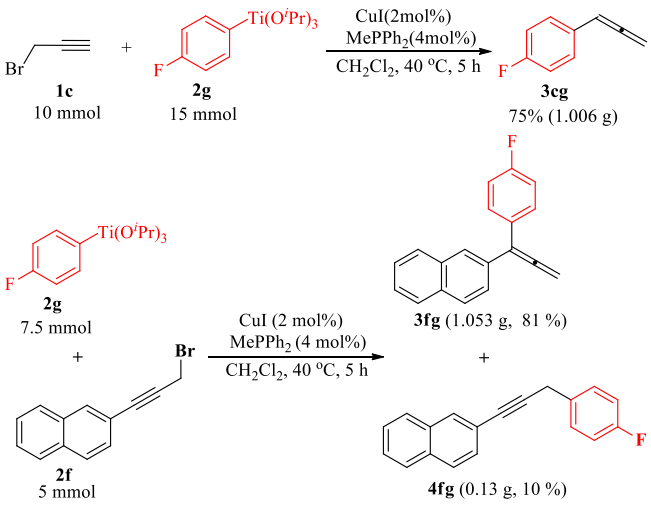
In order to further expand the substrate range of this reaction, we investigated the substitution reaction of propargyl esters (**1g**) with phenyltitanium reagent (**2a**) catalyzed by 2 mol% of CuI/4 mol% $MePPh_2$. However, under optimized reaction conditions, the reaction could not proceed (Scheme 2).

The reaction was found to be effective in gram-scale synthesis, which indicated its potential for practical application (Scheme 3). The desired di-substitute allene products **3 cg** and **3 fg** were synthesized in 1.006 and 1.063 g using this methodology.

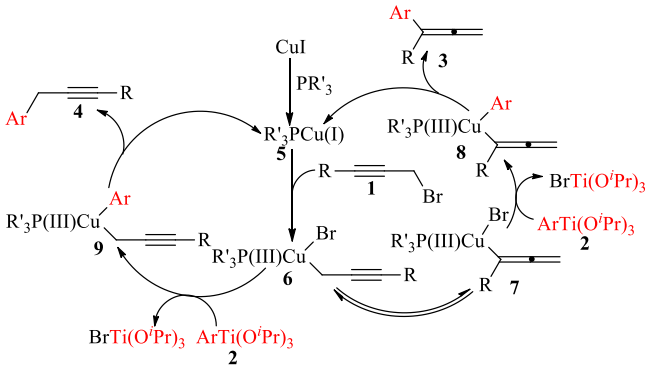
Based on experimental results and relevant literature reports [14], a proposed possible reaction mechanism for the cross-coupling reaction is shown in Scheme 4. The first reaction involves the coordination of PR'$_3$ with CuI to give a active species of Cu(I)PR'$_3$ (**5**). Then, oxidative addition of propargyl bromide to complex **5** gives a Cu(III) species of R'$_3$PCu(CH_2–C≡C–R)Br (**6**). Complex **6** could be isomerized to the corresponding complex **7**. Transmetalation of aryltitanium with **6** or **7** gives aryl (propargyl) copper (III) intermediate **9** or aryl (allenyl)copper (III) intermediate **8** and BrTi(O^{i}Pr)$_3$. Finally complex **9** or **8** undergoes reductive elimination giving the desired product of an allene **3** or an alkyne **4**, and regenerates the active R'$_3$PCu(I) species for the next catalytic cycle.



Scheme 2. The substitution reaction of propargyl ester (**1g**) with phenyltitanium reagent (**2a**).



Scheme 3. Preparative scale synthesis of selected compounds.



Scheme 4. The proposed catalytic cycle for the formation of **3** and **4**.

3. Conclusions

In conclusion, a copper-catalyzed the S_N2' substitute reaction of propargyl bromides with aryltitanium reagents was reported. The S_N2' substitute reactions of aromatic propargyl bromides and aliphatic propargyl bromides with aryltitanium afforded di-substituted allenes in good yields (39–95 %) with high selectivities (up to 99:1). The S_N2' substitute reactions of aromatic propargyl bromides **1** (**d**,**e**,**f**) producing the di-substituted allenes of **3** (**da-fh**) in 39–83 % isolated yields with selectivities of up to 99 %. The S_N2' substitute reactions of aliphatic propargyl bromides **1** (**a**,**b**,**c**) with aryltitaniums can proceeded smoothly to give the di-substituted allenes of **3** (**aa-cg**) in excellent yields (up to 95 %) with high selectivities (up to 99 %). More importantly, the reaction was found to be effective in gram-scale synthesis. Notably, no other co-catalysts and additives are necessary in the present procedure. This methodology provides useful procedure for the synthesis of di-substituted allenes. Further studies on the application of this catalyst to other organotitanium reagents are currently under way.

4. Experimental section

4.1. General procedures

^{1}H NMR and ^{13}CNMR spectra were recorded on a Varian 400 MHz spectrometer. The chemical shifts are reported relative to TMS. Analytical thin-layer chromatography (TLC) was performed on silica 60F-254 plates. Flash column chromatography was carried out on silica

gel (200–400 mesh). All synthesis and manipulations were carried out under a dry nitrogen atmosphere. Chemical reagents and solvents were purchased from Damas-beta and Aldrich, and were used without further purification with the exception of these reagents: THF, Et_2O, Hexane and Toluene were distilled from Sodium under Nitrogen, and dichloromethane was distilled from CaH_2. Organotitanium compounds of RTi(O^iPr)$_3$ (R = Ph (**2a**), 4-$CH_3C_6H_4$ (**2b**), 4-$CH_3OC_6H_4$ (**2c**), 3-$CH_3OC_6H_4$ (**2d**), 2-$CH_3OC_6H_4$ (**2e**), 4-FC_6H_4 (**2f**), 3-FC_6H_4 (**2g**), or 2-naphthyl (**2h**)) were prepared according to literature procedures[6g,11]. Compounds of propargyl halides **1d-1f** were prepared according to literature procedures [14]. Compound of propargyl **1g** was prepared according to literature procedures [6f]. Purification of the reaction products was carried out by flash chromatography.

4.2. General procedures for the synthesis of organotitanium reagents (2a–2h)

To a three-necked round bottom flask containing magnesium turnings (2.43 g, 0.100 mol) in 100 mL of THF and equipped with an addition funnel, a septum and a condenser, aryl bromide (0.100 mol) in 50 mL THF was slowly added over a period of 1 h under a dry nitrogen atmosphere. The reaction mixture was stirred for another 2 h to give a Grignard solution. The above solution was transferred via a cannula to a solution of Ti(O^iPr)$_4$ (22.4 mL, 0.0750 mol) and $TiCl_4$ (2.80 mL, 0.0250 mol) in 50 mL THF, cooling to 0 °C. The resultant solution was allowed to warm to room temperature and reacted for 3 h. The solvent was removed under reduced pressure to give a solid. The residue was extracted with hexane (3 × 100 mL), and the combined extract was concentrated and cooled at 4 or −18 °C to give a crystalline material ArTi(O^iPr)$_3$ (**2a–2h**).

4.3. General procedures for the synthesis of propargyl bromides 1 (d-f)

To a solution of the alkyne (10.0 mmol) in anhydrous THF (11.5 mL) at −40 °C under nitrogen atmosphere was added 4.2 mL n-BuLi (2.5 M, 10.5 mmol). The reaction was stirred at this temperature for 1 h. Then, the 0.36 g paraformaldehyde (12.0 mmol) was added and the reaction was stirred at room temperature for 5 h. The reaction mixture was quenched with a saturated aqueous NH_4Cl solution. The mixture was extracted with dichloromethane (3 × 25 mL). The combined organic layers were washed with brine, dried over magnesium sulfate, filtered and evaporated to give the crude product. The crude product was subjected to flash column chromatography on silica gel (hexane or ethyl acetate and hexane) to afford the corresponding propargylic bromides **1 (d–f)**.

4.4. General procedures for the synthesis of propargyl esters 1 g

To a solution of the ethynylbenzene (16.5 mmol) in anhydrous THF (25 mL) at −78 °C under nitrogen atmosphere was added n-BuLi 1.6 M (16.5 mmol). The reaction was stirred at this temperature for 20 min then at room temperature for 1 h. After cooling to −78 °C, the benzaldehyde (15.0 mmol) was added and the reaction was stirred at room temperature for 1 h. After addition of acetate anhydrous (30.0 mmol) at 0 °C, the reaction mixture was warmed to room temperature and stirred for 2 h before quenched with a saturated aqueous NH_4Cl solution. The mixture was extracted with diethyl ether (3 × 30 mL). The combined organic layers were washed with brine, dried over magnesium sulfate, filtered and evaporated to give the crude products. The crude product was subjected to flash column chromatography on silica gel (hexane or ethyl acetate and hexane) to afford the corresponding propargylic esters (**1g**).

4.5. General procedures for the S_N2' substitute reaction of propargyl bromides with aryltitanium reagents

Under a dry nitrogen atmosphere, CuI (3.5 mg, 0.01 mmol), $MePPh_2$ (2.6 mg, 0.02 mmol) and aryltitanium reagents (0.75 mmol) were added in a reaction vessel. Then propargyl bromide (0.50 mmol) and 1 mL CH_2Cl_2 were added. The resulted solution was stirred at 40 °C (oil bath) for 5 h. After completion the reaction, the mixture was diluted with saturated ammonium chloride solution (5 mL) and extracted with ethyl acetate (3 × 15 mL). The combined organic layers were dried over anhydrous Na_2SO_4, filtered and evaporated *in vacuum*. The residue was subjected to flash column chromatography on silica gel (PE or PE: DCM = 5:1) to afford the corresponding allene products **3**.

penta-1,2-dien-3-ylbenzene**(3aa)** [6g]: colorless oil; yield: 0.057 g (79 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.41 (d, J = 7.6 Hz, 2H), 7.31 (t, J = 7.6 Hz, 2H), 7.20 (d, J = 7.2 Hz, 1H), 5.10 (t, J = 3.2 Hz, 2H), 2.44–2.41 (m, 2H), 1.15 (t, J = 7.2 Hz, 3H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 208.50, 136.65, 128.47, 126.67, 126.02, 106.81, 78.90, 22.50, 12.61.

1-methyl-4-(penta-1,2-dien-3-yl)benzene(**3 ab**) [6b]: colorless oil; yield: 0.058 g (73 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.28 (d, J = 8.0 Hz, 1H), 7.10 (d, J = 7.6 Hz, 1H), 5.06 (t, J = 3.2 Hz, 1H), 2.41–2.37 (m, 1H), 2.31 (s, 1H), 1.14 (t, J = 7.2 Hz, 1H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 208.31, 136.30, 133.64, 129.19, 125.91, 106.64, 78.74, 22.55, 21.16, 12.61.

1-methoxy-4-(penta-1,2-dien-3-yl)benzene(**3ac**) [15]: colorless oil; yield: 0.070 g (81 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.23 (d, J = 8.4 Hz, 2H), 6.82–6.75 (m, 2H), 4.98 (s, 2H), 3.68 (s, 3H), 2.31–2.29 (m, 2H), 1.05 (t, J = 7.2 Hz, 3H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 208.12, 158.54, 129.53, 128.86, 127.08, 120.74, 113.93, 106.31, 78.77, 55.34, 22.66, 12.59.

1-methoxy-3-(penta-1,2-dien-3-yl)benzene(**3ad**): colorless oil; yield: 0.075 g (86 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.21 (t, J = 8.0 Hz, 1H), 7.01–6.96 (m, 2H), 6.74–6.72 (m, 1H), 5.08 (t, J = 3.2 Hz, 2H), 3.77 (s, 3H), 2.42–2.37 (m, 2H), 1.14 (t, J = 7.2 Hz, 3H). ^{13}C NMR (101 MHz, $CDCl_3$) δ 208.49, 159.77, 138.17, 129.31, 118.50, 111.97, 111.87, 106.75, 78.90, 55.20, 22.49, 12.57. HRMS (ESI) calcd for $C_{12}H_{15}O^+$ (M + H)$^+$: 175.1123, found: 175.1117.

1-fluoro-4-(penta-1,2-dien-3-yl)benzene(**3af**) [6b]: colorless oil; yield: 0.061 g (75 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.37–7.33 (m, 2H), 6.99 (t, J = 8.8 Hz, 2H), 5.09 (t, J = 3.2 Hz, 2H), 2.41–2.36 (m, 2H), 1.14 (t, J = 7.2 Hz, 3H). ^{13}C NMR (101 MHz, $CDCl_3$) δ 208.30, 161.82 (d, J = 246.4 Hz), 132.56 (d, J = 3.0 Hz), 127.49 (d, J = 8.1 Hz), 115.20 (d, J = 21.2 Hz), 106.02, 79.12, 22.67, 12.52. ^{19}F NMR (376 MHz, $CDCl_3$) δ −116.37.

1-fluoro-3-(penta-1,2-dien-3-yl)benzene(**3 ag**): colorless oil; yield: 0.055 g (68 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.19–7.14 (m, 1H), 7.08 (d, J = 7.6 Hz, 1H), 7.01 (d, J = 10.8 Hz, 1H), 6.79 (t, J = 8.0 Hz, 1H), 5.04 (s, 2H), 2.33–2.27 (m, 2H), 1.06 (t, J = 7.2 Hz, 3H). ^{13}C NMR (101 MHz, $CDCl_3$) δ 208.60, 163.09 (d, J = 215.1 Hz), 139.33 (d, J = 8.1 Hz), 129.75 (d, J = 8.1 Hz), 121.58 (d, J = 2.0 Hz), 113.42 (d, J = 21.2 Hz), 112.90 (d, J = 22.2 Hz), 106.24, 79.38, 22.43, 12.51. ^{19}F NMR (376 MHz, $CDCl_3$) δ −113.78. HRMS (ESI) calcd for $C_{11}H_{11}FK$ (M + K)$^+$: 201.0476, found: 201.0440.

2-(penta-1,2-dien-3-yl)naphthalene (**3ah**) [26]: white solid; yield: 0.079 g (81 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.74–7.66 (m, 4H), 7.54 (d, J = 8.8 Hz, 1H), 7.40–7.31 (m, 2H), 5.11 (t, J = 3.2 Hz, 2H), 2.50–2.44 (m, 2H), 1.13 (t, J = 7.2 Hz, 3H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 209.17, 134.00, 133.75, 132.49, 128.08, 127.85, 127.68, 126.17, 125.69, 125.41, 123.56, 107.10, 79.30, 22.49, 12.66.

buta-2,3-dien-2-ylbenzene**(3ba)** [6g]: colorless oil; yield: 0.049 g (75 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.41 (d, J = 7.6 Hz, 2H), 7.32 (t, J = 7.6 Hz, 2H), 7.19 (t, J = 7.2 Hz, 1H), 5.02 (d, J = 3.2 Hz, 2H), 2.09 (t, J = 3.2 Hz, 3H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 209.08, 136.80, 128.44, 126.68, 125.77, 99.90, 77.04, 16.82.

1-(buta-2,3-dien-2-yl)-4-methylbenzene(**3bb**) [16]: colorless oil; yield: 0.051 g (71 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.29 (d, J = 8.0 Hz,

2H), 7.12 (d, $J = 8.0$ Hz, 2H), 4.99 (d, $J = 2.8$ Hz, 2H), 2.32 (s, 3H), 2.07 (t, $J = 2.8$ Hz, 3H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 208.90, 136.35, 133.82, 129.16, 125.68, 99.74, 76.89, 21.19, 16.87.

1-(buta-2,3-dien-2-yl)-4-methoxybenzene(**3bc**) [16]: colorless oil; yield: 0.062 g (77 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.24–7.19 (m, 2H), 6.82–6.76 (m, 2H), 4.90 (d, $J = 1.6$ Hz, 2H), 3.69 (s, 3H), 1.98 (s, 3H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 208.70, 158.57, 129.54, 129.06, 126.87, 120.74, 113.90, 99.42, 76.90, 55.36, 16.98.

1-(buta-2,3-dien-2-yl)-3-methoxybenzene(**3bd**) [16]: colorless oil; yield: 0.070 g (88 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.23 (t, $J = 8.0$ Hz, 1H), 7.01–6.96 (m, 2H), 6.75 (d, $J = 7.6$ Hz, 1H), 5.01 (d, $J = 2.8$ Hz, 2H), 3.79 (s, 3H), 2.08 (s, 3H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 209.11, 159.77, 138.38, 129.32, 118.32, 112.03, 111.62, 99.85, 77.06, 55.28, 16.83.

1-(buta-2,3-dien-2-yl)-4-fluorobenzene(**3bf**) [17]: colorless oil; yield: 0.058 g (79 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.37–7.33 (m, 2H), 7.00 (t, $J = 8.8$ Hz, 2H), 5.01 (d, $J = 2.8$ Hz, 2H), 2.07 (t, $J = 3.2$ Hz, 3H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 208.87, 161.86 (d, $J = 247.5$ Hz), 132.73, 127.28 (d, $J = 8.1$ Hz), 115.28 (d, $J = 21.2$ Hz), 99.12, 77.24, 16.98. ^{19}F NMR (376 MHz, $CDCl_3$) δ −116.38.

1-(buta-2,3-dien-2-yl)-3-fluorobenzene(**3bg**) [18]: colorless oil; yield: 0.045 g (61 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.20–7.15 (m, 1H), 7.10 (d, $J = 8.0$ Hz, 1H), 7.01 (d, $J = 10.4$ Hz, 1H), 6.78 (d, $J = 8.0$ Hz, 1H), 4.96 (d, $J = 2.0$ Hz, 2H), 1.98 (s, 3H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 209.18, 163.22 (d, $J = 245.4$ Hz), 139.51 (d, $J = 8.1$ Hz), 129.74 (d, $J = 9.1$ Hz), 121.37 (d, $J = 3.0$ Hz), 113.44 (d, $J = 21.2$ Hz), 112.68 (d, $J = 22.2$ Hz), 99.34, 77.49, 16.69. ^{19}F NMR (376 MHz, $CDCl_3$) δ −113.74 to −113.80 (m).

propa-1,2-dien-1-ylbenzene(**3ca**) [6g]: colorless oil; yield: 0.042 g (73 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.30 (d, $J = 4.4$ Hz, 4H), 7.21–7.18 (m, 1H), 6.16 (t, $J = 6.8$ Hz, 1H), 5.14 (d, $J = 6.8$ Hz, 2H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 208.96, 133.09, 127.80, 126.08, 125.87, 93.14, 78.00.

1-methyl-4-(propa-1,2-dien-1-yl)benzene(**3 cb**) [6b]: colorless oil; yield: 0.041 g (63 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.18 (d, $J = 7.6$ Hz, 2H), 7.10 (d, $J = 7.6$ Hz, 2H), 6.13 (t, $J = 6.8$ Hz, 1H), 5.11 (d, $J = 6.8$ Hz, 2H), 2.31 (s, 3H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 209.71, 136.75, 130.99, 129.46, 126.70, 93.86, 78.80, 21.30.

1-methoxy-4-(propa-1,2-dien-1-yl)benzene(**3 cc**) [6g]: colorless oil; yield: 0.058 g (79 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.21 (d, $J = 8.0$ Hz, 2H), 6.84 (d, $J = 8.4$ Hz, 2H), 6.12 (t, $J = 6.8$ Hz, 1H), 5.11 (d, $J = 6.8$ Hz, 2H), 3.78 (s, 3H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 209.45, 158.81, 127.86, 126.22, 114.25, 93.47, 78.87, 55.39.

1-methoxy-3-(propa-1,2-dien-1-yl)benzene(**3cd**) [6b]: colorless oil; yield: 0.069 g (95 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.19 (t, $J = 8.0$ Hz, 1H), 6.88–6.84 (m, 2H), 6.73 (d, $J = 8.0$ Hz, 1H), 6.12 (t, $J = 6.8$ Hz, 1H), 5.12 (d, $J = 6.8$ Hz, 2H), 3.76 (s, 3H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 209.89, 159.96, 135.45, 129.63, 119.45, 112.81, 111.91, 94.05, 78.95, 55.20.

1-methoxy-2-(propa-1,2-dien-1-yl)benzene(**3ce**) [6g]: colorless oil; yield: 0.062 g (85 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.30 (d, $J = 7.2$ Hz, 1H), 7.06 (t, $J = 7.6$ Hz, 1H), 6.82 (t, $J = 8.8$ Hz, 1H), 6.74 (d, $J = 8.0$ Hz, 1H), 6.49 (t, $J = 6.8$ Hz, 1H), 5.00 (d, $J = 6.8$ Hz, 2H), 3.71 (s, 3H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 210.26, 155.96, 128.04, 127.83, 122.44, 120.87, 111.01, 87.91, 78.11, 55.61.

1-fluoro-4-(propa-1,2-dien-1-yl)benzene(**3cf**) [6b]: colorless oil; yield: 0.058 g (87 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.27–7.23 (m, 2H), 6.99 (t, $J = 8.4$ Hz, 2H), 6.13 (t, $J = 6.8$ Hz, 1H), 5.14 (d, $J = 6.8$ Hz, 2H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 209.69, 162.00 (d, $J = 264.4$ Hz), 129.94, 128.22(d, $J = 8.1$ Hz), 115.68(d, $J = 22.2$ Hz), 93.15, 79.19. ^{19}F NMR (376 MHz, $CDCl_3$) δ −115.52.

1-fluoro-3-(propa-1,2-dien-1-yl)benzene(**3 cg**) [19]: colorless oil; yield: 0.040 g (60 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.18–7.13 (m, 1H), 6.96–6.90 (m, 2H), 6.79 (t, $J = 8.4$ Hz, 6.04 (t, $J = 6.8$ Hz, 1H), 5.08 (d, $J = 6.8$ Hz, 2H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 210.07, 164.30 (d, $J = 246.4$ Hz), 136.61 (d, $J = 8.1$ Hz), 130.10 (d, $J = 9.1$ Hz), 122.57 (d, $J = 3.0$ Hz), 113.89 (d, $J = 23.2$ Hz), 113.36 (d, $J = 22.2$ Hz), 93.48 (d, $J = 2.0$ Hz), 79.34. ^{19}F NMR (376 MHz, $CDCl_3$) δ −113.59 (d).

2-(propa-1,2-dien-1-yl)naphthalene (**3ch**) [6b]: white solid; yield: 0.065 g (78 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.69 (t, $J = 6.4$ Hz, 3H), 7.56 (s, 1H), 7.41 (d, $J = 8.4$ Hz, 1H), 7.37–7.33 (m, 2H), 6.25 (t, $J = 6.8$ Hz, 1H), 5.13 (d, $J = 6.8$ Hz, 2H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 210.45, 133.81, 132.73, 131.54, 128.37, 127.84, 127.79, 126.35, 125.75, 125.54, 124.75, 94.47, 79.23.

propa-1,2-diene-1,1-diyldibenzene(**3da**) [16]: yellow oil; yield: 0.069 g (72 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.36–7.30 (m, 8H), 7.25 (t, $J = 6.8$ Hz, 2H), 5.24 (s, 2H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 209.99, 136.35, 128.53, 127.33, 109.28, 78.19.

1-methoxy-4-(1-phenylpropa-1,2-dien-1-yl)benzene(**3dc**) [20]: yellow oil; yield: 0.075 g (68 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.27–7.16 (m, 7H), 6.78 (d, $J = 8.0$ Hz, 2H), 5.13 (s, 2H), 3.69 (s, 3H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 209.77, 159.03, 136.66, 129.66, 128.47, 127.26, 113.99, 112.56, 108.80, 78.01, 55.39.

1-methoxy-3-(1-phenylpropa-1,2-dien-1-yl)benzene(**3dd**): yellow oil; yield: 0.075 g (68 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.29–7.23 (m, 4H), 7.18–7.14 (m, 2H), 6.87–6.83 (m, 2H), 6.74 (d, $J = 8.4$ Hz, 1H), 5.17 (s, 2H), 3.69 (s, 3H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 209.92, 159.79, 137.84, 136.25, 129.46, 128.55, 128.52, 127.35, 121.08, 114.14, 112.92, 109.22, 78.25, 55.35. HRMS (ESI) calcd for $C_{16}H_{15}O^+$ $(M + H)^+$: 223.1117, found: 223.1119.

1-methoxy-2-(1-phenylpropa-1,2-dien-1-yl)benzene(**3de**): yellow oil; yield: 0.043 g (39 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.25–7.16 (m, 6H), 7.08 (s, 1H), 6.91–6.83 (m, 2H), 5.09 (s, 2H), 3.59 (s, 3H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 209.46, 157.36, 136.38, 131.58, 129.27, 128.26, 126.69, 126.65, 125.19, 120.91, 111.51, 105.39, 77.58, 55.72. HRMS (ESI) calcd for $C_{16}H_{15}O^+$ $(M + H)^+$: 223.1117, found:223.1117.

1-fluoro-4-(1-phenylpropa-1,2-dien-1-yl)benzene(**3df**) [21]: yellow oil; yield: 0.077 g (73 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.33–7.25 (m, 7H), 7.01 (t, $J = 8.4$ Hz, 2H), 5.24 (s, 2H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 209.82, 162.21 (d, $J = 247.5$ Hz), 136.22, 132.30(d, $J = 3.0$ Hz), 130.13 (d, $J = 8.1$ Hz), 128.60, 128.41, 127.47, 115.44 (d, $J = 22.2$ Hz), 108.45, 78.36. ^{19}F NMR (376 MHz, $CDCl_3$) δ −115.07.

1-fluoro-3-(1-phenylpropa-1,2-dien-1-yl)benzene(**3 dg**): yellow oil; yield: 0.083 g (79 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.22 (d, $J = 4.4$ Hz, 4H), 7.18–7.11 (m, 2H), 7.01 (d, $J = 7.6$ Hz, 1H), 6.95 (d, $J = 10.4$ Hz, 1H), 6.82 (t, $J = 8.4$ Hz, 1H), 5.14 (s, 2H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 209.98, 163.06 (d, $J = 246.4$ Hz), 138.86 (d, $J = 8.1$ Hz),135.78, 130.54, 129.89 (d, $J = 9.1$ Hz), 128.61 (d, $J = 8.1$ Hz), 127.59, 124.11 (d, $J = 3.0$ Hz), 115.32 (d, $J = 22.2$ Hz), 114.16 (d, $J = 21.2$ Hz), 108.64, 78.61. ^{19}F NMR (376 MHz, $CDCl_3$) δ −113.12 to −113.17 (m). HRMS (ESI) calcd for $C_{15}H_{12}F^+$ $(M + H)^+$: 211.0918, found: 211.0926.

1-methyl-4-(1-phenylpropa-1,2-dien-1-yl)benzene(**3ea**) [20]: yellow oil; yield: 0.048 g (47 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.32–7.26 (m, 3H), 7.34–7.25 (m, 3H), 7.20 (d, $J = 8.0$ Hz, 2H), 5.28 (s, 2H), 2.40 (s, 3H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 209.90, 137.11, 136.56, 133.38, 129.26, 128.52, 128.49, 128.45, 127.27, 109.11, 78.05, 21.29.

1-methoxy-4-(1-(*p*-tolyl)propa-1,2-dien-1-yl)benzene(**3ec**): yellow oil; yield: 0.081 g (69 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.20–7.15 (m, 4H), 7.05 (d, $J = 8.0$ Hz, 2H), 6.78 (d, $J = 8.4$ Hz, 2H), 5.12 (s, 2H), 3.70 (s, 3H), 2.26 (s, 3H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 209.67, 158.98, 137.02, 133.66, 129.64, 129.22, 128.71, 128.38, 113.96, 108.62, 77.87, 55.40, 21.27. HRMS (ESI) calcd for $C_{17}H_{17}O^+$ $(M + H)^+$: 237.1274, found: 237.1274.

1-fluoro-4-(1-(*p*-tolyl)propa-1,2-dien-1-yl)benzene(**3ef**): yellow oil; yield: 0.044 g (39 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.23–7.20 (m, 2H), 7.13 (d, $J = 7.6$ Hz, 2H), 7.05 (d, $J = 8.0$ Hz, 2H), 6.92 (t, $J = 8.4$ Hz, 2H), 5.13 (s, 2H), 2.26 (s, 3H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 209.62, 162.09 (d, $J = 247.5$ Hz), 137.17, 133.14, 132.40(d, $J = 3.0$ Hz), 130.00 (d, $J = 7.1$ Hz), 129.23, 128.23, 115.29 (d, $J = 22.2$ Hz), 108.19, 78.11, 21.17. ^{19}F NMR (376 MHz, $CDCl_3$) δ −115.28. HRMS (ESI) calcd for $C_{16}H_{14}F^+$ $(M + H)^+$: 225.1074, found: 225.1074.

1-fluoro-3-(1-(*p*-tolyl)propa-1,2-dien-1-yl)benzene(**3eg**): yellow oil; yield: 0.071 g (63 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.21–7.13 (m, 3H),

7.08–7.03 (m, 3H), 6.98 (d, J = 10.4 Hz, 1H), 6.85 (t, J = 8.0 Hz, 1H), 5.17 (s, 2H), 2.27 (s, 3H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 209.87, 163.05 (d, J = 246.4 Hz), 139.10, 137.41, 132.78, 129.83 (d, J = 8.1 Hz), 129.37, 128.47, 124.08 (d, J = 2.0 Hz), 115.29 (d, J = 22.2 Hz), 114.09 (d, J = 21.2 Hz), 108.49, 78.48, 21.29. ^{19}F NMR (376 MHz, $CDCl_3$) δ −113.40. HRMS (ESI) calcd for $C_{16}H_{14}F^+$ $(M + H)^+$: 225.1074, found: 225.1080.

2-(1-phenylpropa-1,2-dien-1-yl)naphthalene (**3fa**) [22]: white solid; yield: 0.054 g (45 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.69 (d, J = 8.0 Hz, 2H), 7.65 (s, 2H), 7.43 (d, J = 8.4 Hz, 1H), 7.32 (d, J = 6.4 Hz, 4H), 7.26 (t, J = 6.8 Hz, 2H), 7.20 (d, J = 6.8 Hz, 1H), 5.20 (s, 2H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 210.41, 136.41, 133.76, 133.59, 132.76, 128.72, 128.61, 128.10, 127.73, 127.46, 127.16, 126.70, 126.27, 126.04, 109.45, 78.39.

2-(1-(4-methoxyphenyl)propa-1,2-dien-1-yl)naphthalene (**3fc**): yellow solid; yield: 0.058 g (43 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.73–7.66 (m, 4H), 7.45 (d, J = 8.4 Hz, 1H), 7.36 (d, J = 2.8 Hz, 2H), 7.25 (d, J = 8.0 Hz, 2H), 6.83 (d, J = 8.0 Hz, 2H), 5.21 (s, 2H), 3.73 (s, 3H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 210.23, 159.14, 134.09, 133.60, 132.76, 129.87, 128.57, 128.10, 128.06, 127.73, 127.06, 126.70, 126.25, 126.00, 114.09, 108.97, 78.22, 55.45. HRMS (ESI) calcd for $C_{20}H_{17}O^+$ $(M + H)^+$: 273.1274, found: 273.1276.

2-(1-(3-methoxyphenyl)propa-1,2-dien-1-yl)naphthalene (**3fd**): yellow solid; yield: 0.094 g (69 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.72–7.65 (m, 4H), 7.44 (d, J = 8.4 Hz, 1H), 7.36–7.33 (m, 2H), 7.19 (t, J = 7.6 Hz, 1H), 6.92–6.88 (m, 2H), 6.77 (d, J = 8.4 Hz, 1H), 5.22 (s, 2H), 3.68 (s, 3H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 210.31, 159.86, 137.88, 133.66, 133.58, 132.76, 129.55, 128.12, 128.08, 127.72, 127.20, 126.69, 126.27, 126.05, 121.25, 114.25, 113.12, 109.39, 78.45, 55.37. HRMS (ESI) calcd for $C_{20}H_{17}O^+$ $(M + H)^+$: 273.1274, found: 273.1277.

2-(1-(2-methoxyphenyl)propa-1,2-dien-1-yl)naphthalene (**3fe**): yellow oil; yield: 0.063 g (46 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.64 (d, J = 8.4 Hz, 2H), 7.57–7.55 (m, 1H), 7.45 (d, J = 8.8 Hz, 1H), 7.40 (s, 1H), 7.28–7.22 (m, 4H), 6.92 (t, J = 7.6 Hz, 1H), 6.85 (d, J = 8.0 Hz, 1H), 5.15 (s, 2H), 3.55 (s, 3H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 209.99, 157.44, 133.83, 133.64, 132.53, 131.68, 129.38, 128.07, 127.78, 127.64, 126.00, 125.61, 125.32, 125.16, 120.97, 111.57, 105.66, 77.85, 55.73. HRMS (ESI) calcd for $C_{20}H_{17}O^+$ $(M + H)^+$: 273.1274, found: 273.1275.

2-(1-(4-fluorophenyl)propa-1,2-dien-1-yl)naphthalene (**3ff**): yellow oil; yield: 0.108 g (83 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.69–7.60 (m, 4H), 7.40–7.22 (m, 5H), 6.93 (t, J = 8.0 Hz, 2H), 5.18 (s, 2H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 210.23, 162.29(d, J = 248.5 Hz), 133.59(d, J = 6.1 Hz), 132.79, 132.33, 130.31(d, J = 8.1 Hz), 128.15(d, J = 7.1 Hz), 127.74, 127.06, 126.51, 126.36, 126.14, 115.53 (d, J = 22.2 Hz), 108.61, 78.54. ^{19}F NMR (376 MHz, $CDCl_3$) δ −114.83. HRMS (ESI) calcd for $C_{19}H_{14}F^+$ $(M + H)^+$: 261.1074, found: 261.1089.

2-(1-(3-fluorophenyl)propa-1,2-dien-1-yl)naphthalene (**3 fg**): yellow oil; yield: 0.086 g (66 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.69–7.62 (m, 2H), 7.38–7.30 (m, 1H), 7.19–7.14 (m, 1H), 7.06–7.00 (m, 1H), 6.86 (t, J = 7.6 Hz, 1H), 5.20 (s, 1H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 210.35, 163.09 (d, J = 246.4 Hz), 138.92, 138.84, 133.56, 133.15, 132.83, 129.97 (d, J = 8.1 Hz), 128.19 (d, J = 16.2 Hz), 127.75, 127.27, 126.58, 126.39, 126.21, 124.28 (d, J = 2.0 Hz), 115.51 (d, J = 21.2 Hz), 114.31 (d, J = 21.2 Hz), 108.80, 78.81. ^{19}F NMR (376 MHz, $CDCl_3$) δ −112.95 to −113.01 (m). HRMS (ESI) calcd for $C_{19}H_{14}F^+$ $(M + H)^+$: 261.1074, found: 261.1077.

2,2'-(propa-1,2-diene-1,1-diyl)dinaphthalene (**3fh**): yellow oil; yield: 0.095 g (65 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.74–7.70 (m, 6H), 7.64 (d, J = 5.2 Hz, 2H), 7.48 (d, J = 8.8 Hz, 2H), 7.35–7.33 (m, 4H), 5.26 (s, 2H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 210.77, 133.85, 133.63, 132.83, 128.19, 128.14, 127.77, 127.39, 126.84, 126.32, 126.12, 109.59, 78.58. HRMS (ESI) calcd for $C_{23}H_{17}^+$ $(M + H)^+$: 293.1325, found: 293.1330.

prop-1-yne-1,3-diyldibenzene(**4da**) [23]: yellow oil; yield: 0.024 g (25 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.46–7.40 (m, 4H), 7.35–7.24 (m, 6H), 3.83 (s, 2H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 136.87, 131.76, 128.67, 128.36, 128.08, 127.94, 126.76, 123.79, 87.65, 82.79, 25.87.

1-fluoro-4-(3-phenylprop-2-yn-1-yl)benzene(**4df**) [23]: yellow oil; yield: 0.019 g (18 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.45–7.43 (m, 2H), 7.38–7.35 (m, 2H), 7.30–7.29 (m, 3H), 7.02 (t, J = 8.8 Hz, 2H), 3.79 (s, 2H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 161.72 (d, J = 245.4 Hz), 132.37(d, J = 3.0 Hz), 131.62, 129.38(d, J = 8.1 Hz), 128.34(d, J = 16.2 Hz), 127.93, 123.48, 115.32 (d, J = 21.2 Hz), 87.27, 82.82, 25.01. ^{19}F NMR (376 MHz, $CDCl_3$) δ −116.57.

1-fluoro-3-(3-phenylprop-1-yn-1-yl)benzene(**4 dg**) [27]: yellow oil; yield: 0.014 g (13 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.38 (d, J = 4.4 Hz, 2H), 7.23–7.17 (m, 4H), 7.09 (t, J = 3.2 Hz, 2H), 6.87 (t, J = 8.0 Hz, 1H), 3.76 (s, 2H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 163.12 (d, J = 247.5 Hz), 150.38, 139.37, 131.79, 130.07 (d, J = 9.1 Hz), 128.27 (d, J = 28.3 Hz), 123.72 (d, J = 3.0 Hz), 123.54, 115.13 (d, J = 22.2 Hz),113.72 (d, J = 21.2 Hz), 86.72, 83.24, 25.65. ^{19}F NMR (376 MHz, $CDCl_3$) δ −113.20.

1-methyl-4-(3-phenylprop-2-yn-1-yl)benzene(**4ea**) [24]: yellow oil; yield: 0.024 g (23 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.33 (d, J = 7.6 Hz, 2H), 7.25 (t, J = 8.0 Hz, 4H), 7.18–7.14 (m, 1H), 7.01 (d, J = 8.0 Hz, 2H), 3.74 (s, 2H), 2.25 (s, 3H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 137.94, 137.02, 131.63, 129.11, 128.64, 128.08, 126.71, 120.71, 86.83, 82.85, 25.89, 21.56.

1-fluoro-4-(3-(*p*-tolyl)prop-2-yn-1-yl)benzene(**4ef**) [28]: yellow oil; yield: 0.022 g (20 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.30–7.25 (m, 3H), 7.02 (d, J = 7.8 Hz, 2H), 6.93 (t, J = 8.8 Hz, 2H), 3.70 (s, 2H), 2.26 (s, 3H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 161.84 (d, J = 245.4 Hz), 138.09, 132.64, 131.63, 129.51 (d, J = 8.1 Hz), 129.15, 120.54, 115.42 (d, J = 21.2 Hz), 86.60, 83.02, 25.16, 21.56. ^{19}F NMR (376 MHz, $CDCl_3$) δ −116.69.

1-fluoro-3-(3-(*p*-tolyl)prop-2-yn-1-yl)benzene(**4eg**): yellow oil; yield: 0.022 g (20 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.27 (d, J = 8.0 Hz, 1H), 7.23–7.16 (m, 1H), 7.10–7.02 (m, 1H), 6.86 (t, J = 8.4 Hz, 1H), 3.74 (s, 1H), 2.26 (s, 1H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 163.12 (d, J = 246.4 Hz), 139.53, 138.16, 131.65, 130.02 (d, J = 8.1 Hz),129.16, 123.71 (d, J = 246.4 Hz), 120.45, 115.12 (d, J = 22.2 Hz), 113.64 (d, J = 21.2 Hz), 85.92, 83.31, 25.66, 21.57. ^{19}F NMR (376 MHz, $CDCl_3$) δ −113.26. HRMS (ESI) calcd for $C_{16}H_{14}F^+$ $(M + H)^+$: 225.1074, found: 225.1069.

2-(3-phenylprop-1-yn-1-yl)naphthalene (**4fa**) [25]: white solid; yield: 0.047 g (39 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.87 (s, 1H), 7.77–7.61 (m, 3H), 7.43–7.36 (m, 5H), 7.27 (t, J = 7.2 Hz, 2H), 7.18 (d, J = 6.8 Hz, 1H), 3.79 (s, 2H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 136.87, 133.14, 132.72, 131.39, 128.78, 128.71, 128.13, 127.99, 127.84, 127.75, 126.80, 126.54, 126.52, 121.11, 88.05, 83.12, 26.00.

2-(3-(4-fluorophenyl)prop-1-yn-1-yl)naphthalene (**4ff**): white solid; yield: 0.013 g (10 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.86 (s, 1H), 7.69–7.64 (m, 3H), 7.40–7.27 (m, 5H), 6.93 (t, J = 8.4 Hz, 2H), 3.73 (s, 2H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 161.88 (d, J = 245.4 Hz), 133.13, 132.77, 132.51, 131.42, 129.55 (d, J = 8.1 Hz), 128.70, 128.04, 127.85, 127.75, 126.59, 120.93, 115.48 (d, J = 21.2 Hz), 87.79, 83.30, 25.24. ^{19}F NMR (376 MHz, $CDCl_3$) δ −116.42. HRMS (ESI) calcd for $C_{19}H_{14}F^+$ $(M + H)^+$: 261.1074, found: 261.1074.

2-(3-(3-fluorophenyl)prop-1-yn-1-yl)naphthalene (**4 fg**): yellow solid; yield: 0.023 g (18 %). ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.02 (s, 1H), 7.83 (t, J = 8.8 Hz, 3H), 7.56–7.51 (m, 3H), 7.38–7.33 (m, 1H), 7.28–7.23 (m, 2H), 7.01 (t, J = 8.4 Hz, 1H), 3.92 (s, 2H). ^{13}C NMR (101 MHz,$CDCl_3$) δ 163.15 (d, J = 246.4 Hz), 139.43, 133.13, 132.81, 131.49, 130.11 (d, J = 9.1 Hz), 128.71, 128.06, 127.82 (d, J = 8.1 Hz),126.61, 123.76 (d, J = 3.0 Hz), 120.83, 115.17 (d, J = 22.2 Hz), 113.76 (d, J = 21.2 Hz), 87.09, 83.57, 25.75. ^{19}F NMR (376 MHz, $CDCl_3$) δ −113.09, −113.11 (d). HRMS (ESI) calcd for $C_{19}H_{14}F^+$ $(M + H)^+$: 261.1074, found: 261.1074.

CRediT authorship contribution statement

Chuan Wu: Resources, Investigation, Formal analysis, Data curation. **Li-Rong Han:** Writing – original draft, Software, Resources, Formal analysis. **Jin-Song Hou:** Software, Resources, Investigation, Data

curation. **Qing-Han Li:** Writing – original draft, Methodology, Funding acquisition, Writing – review & editing, Methodology, Funding acquisition.

Declaration of competing interest

There are no conflicts to declare.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.tet.2024.134345.

Data availability

The data that has been used is confidential.

References

[1] For recent monographs, see. a) N. Krause, A.S.K. Hashmi (Eds.), Modern Allene Chemistry, Wiley-VCH, Weinheim, 2004 vols. 1 and 2;
b) S.M. Ma, Palladium-catalyzed two- or three-component cyclization of functionalized allenes, in: J. Tsuji (Ed.), Palladium in Organic Synthesis, Springer, Berlin, 2005, pp. 183–210.

[2] (a) S.M. Ma, Electrophilic addition and cyclization reactions of allenes, Acc. Chem. Res. 42 (2009) 1679–1688;
(b) J.T. Ye, S.M. Ma, Palladium-catalyzed cyclization reactions of allenes in the presence of unsaturated carbon–carbon bonds, Acc. Chem. Res. 47 (2014) 989–1000;
(c) M.P. Muñoz, Silver and platinum-catalysed addition of O–H and N–H bonds to allenes, Chem. Soc. Rev. 43 (2014) 3164–3183;
(d) J. Lei, Z.G. Xu, D.Y. Tang, Y. Li, J. Xu, H.Y. Li, J. Zhu, Z.Z. Chen, Acid-promoted one-pot synthesis of substituted furan and 6-methylpyrazin-2(1H)-one derivatives via allene intermediate formed in situ, ACS Comb. Sci. 20 (2018) 292–297;
(e) T.K. Meyer, J.C.A. Oliveira, S.C. Sau, N.W.J. Ang, L. Ackermann, Electrooxidative allene annulations by mild cobalt-catalyzed C–H activation, ACS Catal. 8 (2018) 9140–9147;
(f) M.R. Rodríguez, M. Besora, F. Molina, F. Maseras, M.M. Díaz-Requejo, P. J. Pérez, Intermolecular allene functionalization by silver-nitrene catalysis, J. Am. Chem. Soc. 142 (2020) 13062–13071;
(g) X. Xu, X.Z. Zheng, X. Xu, Synthesis of tetrahydroquinolines by scandium-catalyzed [3+3] annulation of anilines with allenes and dienes, ACS Catal. 11 (2021) 14995–15003;
(h) J. Singh, A. Sharma, A. Sharma, Visible light mediated functionalization of allenes, Org. Chem. Front. 8 (2021) 5651–5667.

[3] (a) A. Hoffmann-Röder, N. Krause, Synthesis and properties of allenic natural products and pharmaceuticals, Angew. Chem. Int. Ed. 43 (2004) 1196–1216;
(b) H.K. Maliszewska, C.A. Valle, Y. Xia, M.J. Marín, Z.A.E. Waller, M.P. Muñoz, Precious metal complexes of bis (pyridyl)allenes: synthesis and catalytic and medicinal applications, Dalton Trans. 50 (2021) 16739–16750.

[4] (a) P. Rivera-Fuentes, F. Diederich, Allenes in molecular materials, Angew. Chem. Int. Ed. 51 (2012) 2818–2828. Angew. Chem. 2012, 124, 2872–2882;
(b) N.J. Galan, J.N. Brantley, General access to allene-containing polymers using the Skattebøl Rearrangement, ACS Macro Lett. 9 (2020) 1662–1666;
(c) M. Macernis, S. Streckaite, R. Litvin, A.A. Pascal, M.J. Llansola-Portoles, B. Robert, L. Valkunas, Electronic and vibrational properties of allene carotenoids, J. Phys. Chem. A 126 (6) (2022) 813–824.

[5] For reviews on the synthesis of allenes, see. (a) L.K. Sydnes, Allenes from cyclopropanes and their use in organic SynthesisRecent developments, Chem. Rev. 103 (2003) 1133–1150;
(b) K.M. Brummond, J.E. Deforrest, Synthesizing allenes today (1982−2006), Synthesis (2007) 795–818;
(c) N. Krause, V. Belting, C. Deutsch, J. Erdsack, H. Fan, B. Gockel, A. Hoffmann-Röder, N. Morita, F. Volz, Golden opportunities in catalysis, Pure Appl. Chem. 80 (2008) 1063–1069;
(d) M. Ogasawara, Catalytic enantioselective synthesis of axially chiral allenes, Tetrahedron: Asymmetry 20 (2009) 259–271;
(e) S.C. Yu, S.M. Ma, How easy are the syntheses of allenes? Chem. Commun. 47 (2011) 5384–5418;
(f) R.K. Neff, D.E. Frantz, Recent advances in the catalytic syntheses of allenes: a critical assessment, ACS Catal. 4 (2014) 519–528.

[6] For selected recent reports on the synthesis of allenes, see. (a) B. Bolte, Y. Odabachian, F. Gagosz, Gold(I)-Catalyzed rearrangement of propargyl benzyl ethers: a practical method for the generation and in situ transformation of substituted allenes, J. Am. Chem. Soc. 132 (2010) 7294–7296;
(b) Q.H. Li, H.M. Gau, Synthesis of allenes via nickel-catalyzed cross-coupling reaction of propargylic bromides with Grignard reagents, Synlett 23 (2012) 747–750;
(c) M.L. Hossain, F. Ye, Y. Zhang, J.B. Wang, CuI-catalyzed cross-coupling of N-tosylhydrazones with terminal alkynes: synthesis of 1,3-disubstituted allenes, J. Org. Chem. 78 (2013) 1236–1241;
(d) T. Hashimoto, K. Sakata, F. Tamakuni, M.J. Dutton, K. Maruoka, Phase-transfer-catalysed asymmetric synthesis of tetrasubstituted allenes, Nat. Chem. 5 (2013) 240–244;
(e) Z. Wu, F. Berhal, M.M. Zhao, Z.G. Zhang, T. Ayad, V. Ratovelomanana-Vidal, Palladium-catalyzed efficient enantioselective synthesis of chiral allenes: steric and electronic effects of ligands, ACS Catal. 4 (2014) 44–48;
(f) Q.H. Li, J.Y. Jeng, H.M. Gau, Highly efficient synthesis of allenes from trimethylaluminum reagent and propargyl acetates mediated by a palladium catalyst, Eur. J. Org. Chem. (2014) 7916–7923;
(g) Q.H. Li, J.W. Liao, Y.L. Huang, R.T. Chiang, H.M. Gau, Nickel-catalyzed substitution reactions of propargyl halides with organotitanium reagents, Org. Biomol. Chem. 12 (2014) 7634–7642;
(h) J. Ruchti, E.M. Carreria, Rh-catalyzed stereospecific synthesis of allenes from propargylic benzoates and arylboronic acids, Org. Lett. 18 (2016) 2174–2176;
(i) M. Periasamy, P.O. Reddy, I. Satyanarayana, L. Mohan, A. Edukondalu, Diastereoselective synthesis of tetrasubstituted propargylamines via hydroamination and metalation of 1-alkynes and their enantioselective conversion to trisubstituted chiral allenes, J. Org. Chem. 81 (2016) 987–999;
(j) Z. Zhang, X.B. Shao, G. Zhang, Q.H. Li, X.Y. Li, Highly efficient synthesis of multi-substituted allenes from propargyl acetates and organoaluminum reagents mediated by palladium, Synthesis 49 (2017) 3643–3653;
(k) Z. Zhang, S. Mo, G. Zhang, X.B. Shao, Q.H. Li, Y. Zhong, Synthesis of allenes via nickel-catalyzed cross-coupling reaction of propargylic bromide with Grignard reagent, Synlett 5 (2017) 611–614;
(l) X.B. Shao, C. Wen, G. Zhang, K.P. Cao, L. Wu, Q.H. Li, Palladium-catalyzed, ligand-free S_N2' substitution reactions of organo-aluminum with propargyl acetates for the synthesis of multisubstituted allenes, J. Organomet. Chem. 870 (2018) 68–75;
(m) X.B. Shao, Z. Zhang, Q.H. Li, Z.G. Zhao, Synthesis of multi-substituted allenes from organoalanes reagents and propargyl esters by using a nickel catalyst, Org. Biomol. Chem. 16 (2018) 4797–4806;
(n) G. Zhang, Y.K. Song, F. Zhang, Z.J. Xue, M.Y. Li, G.S. Zhang, B.B. Zhu, J. Wei, C. Li, C.G. Feng, G.Q. Lin, Palladium-catalyzed allene synthesis enabled by β-hydrogen elimination from sp 2-carbon, Nat. Commun. 12 (2021) 728;
(o) W. Lei, H. Liu, Y. Li, Y.W. Fang, Nickel-catalysed SET-reduction-based access to functionalized allenes via 1,4-carbohydrogenation of 1,3-enynes with alkyl bromides, Org. Chem. Front. 9 (2022) 3862–3868.

[7] (a) S.M. Ma, S.C. Yu, S.H. Yin, Studies on K_2CO_3-catalyzed 1,4-addition of 1,2-allenic ketones with diethyl malonate: controlled selective synthesis of β,γ-unsaturated enones and α-pyrones, J. Org. Chem. 68 (2003) 8996–9002;
(b) M. Yokota, K. Fuchibe, M. Ueda, Y. Mayumi, J. Ichikawa, Facile synthesis of 1,1-difluoroallenes via the difluorovinylidenation of aldehydes and ketones, Org. Lett. 11 (2009) 3994–3997.

[8] (a) M. Brossat, M.P. Heck, C. Mioskowski, Bistrimethylsilyl propargylic ether: a versatile ambident synthon to access substituted allenyne ethers and α-substituted bispropargylic alcohols, J. Org. Chem. 72 (2007) 5938–5941;
(b) N. Phadke, M. Findlater, Formation of iridium(III) allene complexes via isomerization of internal alkynes, Organometallics 33 (2014) 16–18.

[9] (a) J.K. Kuang, S.M. Ma, One-pot synthesis of 1,3-disubstituted allenes from 1-alkynes, aldehydes, and morpholine, J. Am. Chem. Soc. 132 (2010) 1786–1787;
(b) G.J. Jiang, Q.H. Zheng, M. Dou, L.G. Zhuo, W. Meng, Z.X. Yu, Mild-condition synthesis of allenes from alkynes and aldehydes mediated by tetrahydroisoquinoline (THIQ), J. Org. Chem. 78 (2013) 11783–11793;
(c) Q. Liu, X.J. Tang, Y.J. Cai, S.M. Ma, Catalytic one-pot synthesis of trisubstituted allenes from terminal alkynes and ketones, Org. Lett. 19 (2017) 5174–5177;
(d) D.M. Lustosa, S. Clemens, M. Rudolph, A.S.K. Hashmi, Gold-catalyzed one-pot synthesis of 1,3-disubstituted allenes from benzaldehydes and terminal alkynes, Adv. Synth. Catal. 361 (2019) 5050–5056;
(e) L.P. Zorba, E. Egaña, E. Gómez-Bengoa, G.C. Vougioukalakis, Zinc iodide catalyzed synthesis of trisubstituted allenes from terminal alkynes and ketones, ACS Omega 6 (2021) 23329–23346.

[10] (a) C. Deutsch, B.H. Lipshutz, N. Krause, Angew. Chem. Int. Ed. 46 (2007) 1650–1653;
(b) H. Ito, Y. Sasaki, M. Sawamura, Copper(I)-Catalyzed substitution of propargylic carbonates with diboron: selective synthesis of multisubstituted allenylboronates, J. Am. Chem. Soc. 130 (2008) 15774–15775;
(c) Y.L. Wang, W.L. Zhang, S.M. Ma, A room-temperature catalytic asymmetric synthesis of allenes with ECNU-phos, J. Am. Chem. Soc. 135 (2013) 11517–11520;
(d) H.L. Li, D. Müller, L. Guénée, A. Alexakis, Synthesis of multisubstituted pyridines, Org. Lett. 15 (2013) 334–337;
(e) Y.X. Zhang, S. Du, T. Yang, F.Y. Jin, J.Y. Zhou, B.P. Cao, Z.J. Mao, X.R. Song, Q. Xiao, Direct and efficient synthesis of tetrasubstituted allenyl organothiophosphates from propargylic alcohols under catalyst- and additive-free

conditions, Org. Chem. Front. 9 (2022) 3156–3162;
(f) Y.Z. Yang, K.J. Szabó, Synthesis of allenes by catalytic coupling of propargyl carbonates with aryl iodides in the presence of diboron species, J. Org. Chem. 81 (2016) 250–255;
(g) Z. Zhang, M.M. Ji, X.F. Wu, Y.Y. He, J.B. Peng, Synthesis of multisubstituted 2,3-allenamides via palladium-catalyzed carbonylation of propargylic esters, J. Org. Chem. 89 (2024) 9001–9010.

[11] (a) C.R. Chen, S.L. Zhou, D.B. Biradar, H.M. Gau, Adv. Synth. Catal, Extremely Efficient Cross-Coupling of Benzylic Halides with Aryltitanium Tris (isopropoxide) Catalyzed by Low Loadings of a Simple Palladium(II) Acetate/Tris(p-tolyl) phosphine System 352 (2010) 1718–1727;
(b) S.T. Chang, Q.H. Li, R. Chiang, T, H.M. Gau, Palladium-catalyzed coupling reactions of $(ArCH_2)Ti(O\text{-}i\text{-}Pr)_3$ with aromatic or heteroaromatic bromides, Tetrahedron 68 (2012) 3956–3962;
(c) C. Wu, Q.H. Li, Highly efficient synthesis of 1,2-disubstituted acetylenes derivatives from the cross-coupling reactions of 1-bromoalkynes with organotitanium reagents, Tetrahedron 74 (2021) 6063–6070;
(d) Q.H. Li, H.L. Xiao, X.Y. Jia, J.X. Pu, L.R. Han, C. Wu, Recent advances in cross-coupling reactions of trialkoxy titanium reagents $(RTi(OR')_3)$ with electrophilic reagents, Tetrahedron 139 (2023) 133451;
(e) Z.H. Zhang, J.X. Pu, X.Y. Jia, J.S. Hou, L.R. Han, C. Wu, Q.H. Li, Highly efficient synthesis of arylsulfide derivatives from coupling reaction of aryl titanium reagents with N-aryl thiosuccinimides catalyzed by copper under ligand-free conditions, J. Organomet. Chem. (2024), https://doi.org/10.1016/j.jorganchem.2024.123192.

[12] (a) Z.X. Zhang, P.H. Chen, G.S. Liu, Copper-catalyzed radical relay in $C(sp^3)$–H functionalization, Chem. Soc. Rev. 51 (2022) 1640–1658;
(b) L.J. Cheng, .N.P. Mankad, C–C and C–X coupling reactions of unactivated alkyl electrophiles using copper catalysis, Chem. Soc. Rev. 49 (2020) 8036–8064.

[13] (a) S. Mo, X.B. Shao, G. Zhang, Q.H. Li, Highly efficient synthesis of unsymmetrical 1,3-diynes from organoalane reagents and alkynyl bromides mediated by a nickel catalyst, RSC Adv. 7 (2017) 27243–27247;
(b) X.B. Shao, X. Jiang, Q.H. Li, Z.G. Zhao, Highly efficient synthesis of 1,2-disubstituted acetylenes derivatives from the cross-coupling reactions of 1-bromoalkynes with organoalane reagents, Tetrahedron 74 (2018) 6063–6070;
(c) G. Zhang, K. Wu, C. Wen, Q.H. Li, Nickel-catalyzed cross-coupling of organo aluminum reagents with alkynylhalides for the synthesis of symmetrical and unsymmetrical conjugated 1,3-diynes derivatives, J. Organomet. Chem. 906 (2020) 121040, 10.1016/j.jorganchem.2019. 121 040;
(d) C. Wen, X. Jiang, K. Wu, R.Q. Luo, Q.H. Li, Highly efficient synthesis of unsymmetrical 1,3-diynes from organoalanes reagents and alkynyl bromides mediated by nickel catalyst, RSC Adv. 10 (2020) 19610–19614;
(e) C. Wen, C. Wu, R.Q. Luo, Q.H. Li, F. Chen, Highly efficient synthesis of 2-substituted benzo[*b*]furan derivatives from the cross-coupling reactions of 2-Halobenzo[*b*]furans with organoalane reagents, Synthesis 53 (2021) 3847–3861;
(f) H.L. Xiao, G. Zhang, R.Q. Luo, Q.H. Li, Simple and efficient nickel-catalyzed cross-coupling reaction of alkenylalanes with alkynyl halides for synthesis of conjugated enynes, Tetrahedron 103 (2022) 132549;
(g) R.Q. Luo, S.P. Guo, H.L. Xiao, Q.H. Li, Cross-coupling reaction of organoalane reagents with 2-mercaptobenzo-5-membered heterocycles for efficient synthesis of benzo-5-membered heterocycle sulfides, Tetrahedron 103 (2022) 132564;
(h) K. Wu, C. Wu, X.Y. Jia, L. Zhou, Q.H. Li, Highly selective cross-coupling reactions of 1,1-dibromoethylenes with alkynylaluminums for the synthesis of aryl substituted conjugated enediynes and unsymmetrical 1, 3-diynes, RSC Adv. 12 (2022) 13314–13318;
(i) X. Jiang, H.L. Xiao, X.Y. Jia, J.X. Pu, L.R. Han, Q.H. Li, The cross-coupling reaction of organoalane reagents with 2-methyl thiobenzo[*d*]thiazoles *via* C-S bond cleavage catalyzed by nickel, New J. Chem. 47 (2023) 14078–14094;
(j) M. Liang, J.X. Pu, X.Y. Jia, L.R. Han, Q.H. Li, Palladium-catalyzed cross-coupling of 2-Benzo[b]thiophene or 2-Benzo[b]furan aluminum with heteroaryl or alkynyl bromides for the synthesis of 2-hetroaryl or alkynyl benzo[b]thiophene or benzo[b]furan derivatives, Lett. Org. Chem. 21 (2024) 116–123.

[14] I. Manjón-Mata, M.T. Quirós, E. Buñuel, D.J. Cárdenas, Regioselective iron-catalysed cross-coupling reaction of aryl propargylic bromides and aryl Grignard reagents, Adv. Synth. Catal. 362 (2020) 146–151.

[15] T. Sawano, K. Ou, T. Nishimura, T. Hayashi, Cobalt-catalyzed asymmetric addition of silylacetylenes to 1, 1-disubstituted allenes, J. Org. Chem. 78 (2013) 8986–8993.

[16] A.M. Gates, W.L. Santos, Regioselective diboron-mediated semireduction of terminal allenes, Synthesis 51 (2019) 4619–4624.

[17] M.Y. Ngai, E. Skucas, M.J. Krische, Ruthenium catalyzed C− C bond formation via transfer hydrogenation: branch-selective reductive coupling of allenes to paraformaldehyde and higher aldehydes, Org. Lett. 10 (2008) 2705–2708.

[18] H. Jang, B. Jung, A.H. Hoveyda, Catalytic enantioselective protoboration of disubstituted allenes. Access to alkenylboron compounds in high enantiomeric purity, Org. Lett. 16 (2014) 4658–4661.

[19] K. Rafizadeh, K. Yates, Photochemical and thermal hydrations of aromatic allenes. Evidence for allyl and vinyl cation intermediates[J], J. Org. Chem. 49 (1984) 1500–1506.

[20] S. Ma, A. Zhang, Efficient synthesis of 1, 1-diaryl 1, 2-dienes via Pd (0)-catalyzed coupling of aryl halides with allenic/propargylic zinc reagents, J. Org. Chem. 63 (1998) 9601–9604.

[21] M. Al-Masum, Y. Yamamoto, Palladium-catalyzed hydrocarboxylation of allenes, J. Am. Chem. Soc. 120 (1998) 3809–3810.

[22] S. Tabuchi, K. Hirano, M. Miura, Stereospecific Pd-catalyzed intermolecular C (sp^3)–C (sp) cross-coupling of diarylmethyl carbonates and terminal alkynes under base-free conditions, Chem. Eur J. 21 (2015) 16823–16827.

[23] M. Qian, E. Negishi, Palladium-catalyzed cross-coupling reaction of alkynylzincs with benzylic electrophiles, Tetrahedron Lett. 46 (2005) 2927–2930.

[24] D.B. Biradar, H.M. Gau, Simple and efficient nickel-catalyzed cross-coupling reaction of alkynylalanes with benzylic and aryl bromides, Chem. Commun. 47 (2011) 10467–10469.

[25] Y. Zhang, D. Zhang, Cu-Photoredox-catalyzed C (sp)–C (sp^3) coupling of redox-active esters with terminal alkynes, Org. Biomol. Chem. 18 (2020) 4479–4483.

[26] J. Han, P.W. Zhang, H.M. Wang, R.H. Zhang, H.H. Wu, J.L. Zhang, Design and synthesis of WJ-Phos, and application in Cu-catalyzed enantioselective boroacylation of 1,1-disubstituted allenes, ACS Catal. 9 (2019) 6890–6895.

[27] Y. Sun, T. Zhao, H.X. Wang, Y. Pan, L.L. Huang, H.D. Feng, Precision propargylic substitution reaction: Pd-catalyzed Suzuki-Miyaura coupling of nonactivated propargylamines with boronic acids, J. Org. Chem. 89 (2024) 13774–13781.

[28] U. Mitsuhiro, N. Kota, R. Ilhyong, Transition-metal-free Suzuki-Miyaura coupling reaction of arylpropargylic bromides with aryl- and alkenylboronic acids, Synlett 24 (2013) 1683–1686.